

Assignment /

Assignment #: 24

Topic : Files

Level 1: Easy

1. You are developing a program that needs to read and display the contents of a text file. Write a Python function `read_file(file_path)` that reads a file and prints its contents.

Sample test Cases:

Test Cases	Test Case1:	Test Case 2:
Input	example1.txt	example2.txt
File Content	Hello, World!	This is a test file.
Output	Hello, World!	This is a test file.

2. You are developing a logging system where messages need to be written to a log file. Write a Python function `write_to_file(file_path, message)` that writes a message to a file

Sample test Cases:

Test Cases	Test Case1:	Test Case 2:
Input	log1.txt	Log2.txt
Content to write	This is the first log message.	Another log entry
Output	This is the first log message.	Another log entry

3. You are developing a system where new entries need to be appended to an existing file. Write a Python function `append_to_file(file_path, message)` that appends a message to a file.

Sample test Cases:

Test Cases	Test Case1:	Test Case 2:
File content Before	First line.	Entry 1
Input	Second line.	Entry 2
Output	First line. Second line.	Entry 1 Entry 2

Level 2: Medium

1. You are developing a program to update the content of a file by replacing specific text. Write a Python function `find_and_replace(file_path, old_text, new_text)` that finds and replaces all occurrences of `old_text` with `new_text` in the given file.

Sample test Cases:

Test Cases	Test Case1:	Test Case 2:
File content Before	Hello, World!	This is a test file.
Output	Hello, Python!	This is a sample file.

2. You are developing a program to read the contents of a file and store each line as an element in a list. Write a Python function `file_to_list(file_path)` that reads a file and returns a list of its lines.

Sample test Cases:

Test Cases	Test Case1:	Test Case 2:
File Content	Line 1 Line 2 Line 3	Only one line.
Output	['Line 1', 'Line 2', 'Line 3']	['Only one line.']

3. You are developing a program to merge the contents of two files into a single file. Write a Python function `merge_files(file1, file2, output_file)` that reads the contents of two files and writes them into a single output file.

Sample test Cases:

Test Cases	Test Case1:	Test Case 2:
File 1 Content	File 1, Line 1 File 1, Line 2	Part 1, Line 1 Part 1, Line 2
File 2 Content	File 2, Line 1 File 2, Line 2	Part 1, Line 1 Part 1, Line 2
Output	File 1, Line 1 File 1, Line 2 File 2, Line 1 File 2, Line 2	Part 1, Line 1 Part 1, Line 2 Part 2, Line 1 Part 2, Line 2

Level 3: Hard

1. You are developing a program to manage student grades. Each student has a unique ID, name, and a list of grades. Write a Python function `manage_student_grades(input_file, output_file)` that reads student data from an input file, calculates the average grade for each student, and writes the results to an output file in the format `ID, Name, Average_Grade`.

Sample test Cases

Test Cases	Test Case1:	Test Case2:
Input File	101, John Doe, 85, 90, 78	201, Alice Brown, 88, 92, 84
Content	102, Jane Smith, 92, 88, 95 103, Bob Johnson, 70, 80, 85	202, Charlie Davis, 75, 80, 85 203, Eve Miller, 90, 85, 95
Files	Input File = students.txt Output File = averages.txt	Input File = grades.txt Output File = student_averages.txt
Output File	101, John Doe, 84.33	201, Alice Brown, 88.00
Content	102, Jane Smith, 91.67 103, Bob Johnson, 78.33	202, Charlie Davis, 80.00 203, Eve Miller, 90.00

2. You are developing a payroll system to generate salary reports for employees. Write a Python function `generate_salary_report(employee_file, report_file)` that reads employee data (ID, name, hourly wage, hours worked) and writes a salary report (ID, name, total salary) to an output file.

Sample test Cases

Test Cases	Test Case1:	Test Case2:
Input File	001, John Doe, 15.0, 40	101, Alice Brown, 18.0, 45
Content	002, Jane Smith, 20.0, 35 003, Bob Johnson, 25.0, 30	102, Charlie Davis, 22.0, 40 103, Eve Miller, 28.0, 38
Files	Input File = employees.txt	Input File = payroll.tx

	Output File = salaries.txt	Output File = salary_report.txt
Output File	001, John Doe, 600.00	101, Alice Brown, 810.00
Content	002, Jane Smith, 700.00	102, Charlie Davis, 880.00
	003, Bob Johnson, 750.00	103, Eve Miller, 1064.00